

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA: Antibody genes with selectable markers.
Species Used	Hybridomas		

Before the Pulse

Cell Growth Medium	DMEM + 15% Horse Serum + L- glutamine (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Mid- log
Wash Solution	Cold Phosphate Buffered Saline	Pre-pulse Incubation	10 minutes on ice.

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	4°C	Cuvette Gap	0.4 cm
Electroporation Medium*	Phosphate Buffered Saline	Voltage	0.160 kV
Cell Density	10 (7) cells / ml	Field Strength	0.4 kV/cm
Volume of Cells	0.8 ml	Capacitor	960 µF
DNA Concentration	20 µg / ml	Resistor	(Pulse Controller) Ω none
DNA Resuspension Buffer	Phosphate Buffered Saline	Time Constant	15 msec
Volume of DNA	Not given		
After the Pulse			
Outgrowth Medium	DMEM + 15% Fetal Calf Serum + Selector		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C	
Length of Incubation	> 1 week	<i>Immunol.</i> , Jan 15, 1991.
Selection Method or Assay Used	G418 or mycophenolic acid	
Electroporation Efficiency	0.5 to 1 transfectants / µg	
Per Cent Survival	Not given	

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Survey Number

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